

SIMATS ENGINEERING
ASSESSMENT TOOL 2 : Scenario-Based

COURSE NAME: Cloud Computing and Big Data Analytics **COURSE CODE:** CSA1521

COURSE OUTCOME COVERED

CO2: *Analyze virtualization architectures to identify performance and security issues in cloud computing environments. (BL3)*

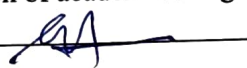
Dave's Taxonomy: Manipulation (Level 2)
Question Count: 5

Total Marks: 25

Code of Conduct

I Karandha (192521215) (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.

I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student: 

Assessment Tasks

Scenario-Based Questions

1. A media company wants to migrate its video streaming platform to the cloud.
The system must handle millions of concurrent users with minimal buffering.
Design a cloud architecture using scalability and caching techniques.
Explain how load balancing and auto-scaling can ensure smooth streaming.
Discuss cost optimization strategies for such high-demand workloads.
2. A fintech startup is deploying applications using containers across multiple cloud providers.
They face challenges in maintaining consistency and managing deployments.
Propose a solution using container orchestration and multi-cloud strategies.
Explain how service discovery and scaling can be handled efficiently.
Highlight the benefits and risks of a multi-cloud deployment model.
3. An e-learning platform experiences data loss due to accidental deletion in the cloud.
The organization needs a robust backup and disaster recovery strategy.
Design a solution using cloud-native backup, replication, and versioning.

Explain how recovery time objective (RTO) and recovery point objective (RPO) are achieved.
Provide steps to ensure business continuity during failures.

4. A government agency is adopting a hybrid cloud model for sensitive and non-sensitive data.
Some applications must remain on-premises due to compliance requirements.
Design a hybrid architecture that ensures secure communication between environments.
Explain how identity management and encryption should be implemented.
Discuss challenges in managing hybrid cloud environments.
5. A SaaS provider needs to ensure high availability and fault tolerance for its application.
The system must continue operating even if one region goes down.
Propose a multi-region deployment strategy in the cloud.
Explain how data replication and failover mechanisms should be configured.
Discuss monitoring and alerting strategies for proactive issue detection.

Scenario-Based Rubric

Criteria	Weightage	Level 4 - Exemplary	Level 3 - Proficient	Level 2 - Developing	Level 1 - Beginning
Understanding of Scenario	20%	Demonstrates complete understanding of the scenario context	Good understanding with minor gaps	Partial understanding	Fails to understand the scenario
Identification of Risks / Issues	20%	Identifies all major issues correctly	Identifies most issues	Limited issue identification	Misses major issues
Application of Concepts	25%	Applies virtualization concepts effectively	Applies concepts with minor mistakes	Partially correct application	Weak application
Decision Justification	20%	Decisions are well justified and technically strong	Reasonable justification	Weak justification	No useful justification
Clarity	15%	Clear and well-organized response	Generally clear	Somewhat unclear	Poorly presented

14 Media company - video streaming platform:

Cloud Architecture:-

- use cloud servers to host videos.
- store videos in cloud storage.
- use CDN (Content Delivery Network) to deliver videos faster.
- use cache to reduce buffering.

Load Balancing and Auto-Scaling:-

- load balancer sends users to different servers.
- Auto scaling adds more servers when traffic increases.
- Removes extra servers when traffic decreases.

Code optimization:-

- use pay-as-you-go pricing.
- store old videos in low-cost storage.

Fintech startup - Multi cloud containers:

Soln:-

- use docker containers.
- use kubernetes for containers orchestration.
- Deploy apps on multiple cloud providers.

Service Discovery and scaling:-

- kubernetes automatically find services.
- it increases or decreases containers based on demand.

Benefits:-

- High availability.
- Better reliability.
- Avoids vendor lock-in.

Risks:-

- more complex management.
- Higher network cost.
- security management difficult.

3/

E-learning platform - Backup and Disaster Recovery.

Soln:-

- Taken regular cloud backups.
- Enable data replication.
- use versioning to recover deleted file.

RTO and RPO:-

- RTO:- Restore the system quickly.
- RPO:- Restore the latest available data with minimum loss.

Business Continuity steps:-

- keeps backups in different region.
- Test Recovery regularly.
- use automatic failover.

4/

Government Agency - Hybrid cloud.

Hybrid Architecture:-

- keep sensitive data on-premises.
- store non-sensitive data in the public cloud.

→ connect both using secure VPN.

Identity Management and Encryption:-

→ use multi-factor authentication (MFA).

→ Encrypt data during storage and transfer.

→ Gives users role-based access.

Challenges:-

→ Complex management.

→ security and compliance issues.

→ network latency.

5) SaaS provider - High Availability.

→ Deploy the apps in multiple cloud regions.

→ If one region fails, another region continues the service.

Data Replication and failover:-

→ copy data to all regions.

→ use automatic failover to switch users to a healthy region.

Monitoring and Alerting:-

→ use cloud monitoring tools to detect issues early.

→ monitor server health continuously.